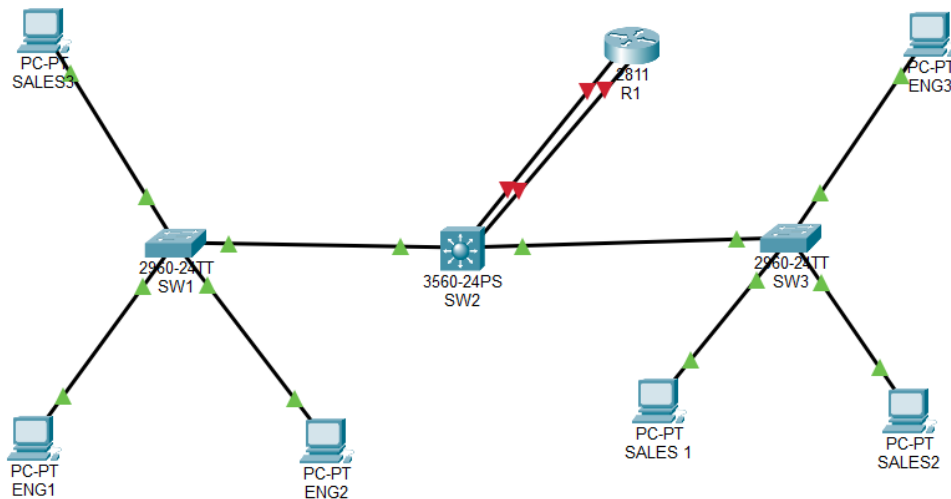


Makata Mpho

Week 9 practical



89.

SW1 Client

```
Switch>ena
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW1
SW1(config)#VTP DOMAIN Campus
Changing VTP domain name from NULL to Campus
SW1(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to Campus.

SW1(config)#VTP Mode Client
Setting device to VTP CLIENT mode.
SW1(config)#VTP password cisco123
Setting device VLAN database password to cisco123
SW1(config)#end
SW1#
```

SW3 Client

```
Switch>
Switch>ena
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW3
SW3(config)#VTP DOMAIN Campus2
Changing VTP domain name from NULL to Campus2
SW3(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to Campus2.

SW3(config)#vtp mode client
Setting device to VTP CLIENT mode.
SW3(config)#vtp password cisco321
Setting device VLAN database password to cisco321
SW3(config)#end
```

SW2

```
Switch#
Switch#ENA
Switch#CONF T
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#HOSTNAME SW2
SW2(config)#VTP DOMAIN Campus
Changing VTP domain name from NULL to Campus
SW2(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to Campus.

SW2(config)#vtp mode Server
Device mode already VTP SERVER.
SW2(config)#vtp password cisco123
Setting device VLAN database password to cisco123
```

90.

```
SW3>
SW3>ENA
SW3>CONF T
Enter configuration commands, one per line. End with CNTL/Z.
SW3(config)#interface f0/1
SW3(config-if)#switchport mode access
SW3(config-if)#switchport access vlan 20
SW3(config-if)#
SW3(config-if)#exit
SW3(config)#interface f0/2
SW3(config-if)#switchport mode access
SW3(config-if)#switchport access vlan 20
SW3(config-if)#
SW3(config-if)#exit
SW3(config)#interface f0/3
SW3(config-if)#switchport mode access
SW3(config-if)#switchport access vlan 10
SW3(config-if)#end
SW3#
%SYS-5-CONFIG_I: Configured from console by console

SW3#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                           Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                           Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                           Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                           Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                           Fa0/24, Gig0/1, Gig0/2
1002 fddi-default          active
1003 token-ring-default     active
1004 fddinet-default        active
1005 trnet-default          active
SW3#
```

```

SW1>ena
SW1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#interface f0/1
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 10
SW1(config-if)#
SW1(config-if)#exit
SW1(config)#interface f0/2
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 10
SW1(config-if)#
SW1(config-if)#exit
SW1(config)#interface f0/3
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 20
SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
1002	fdi-default	active	
1003	token-ring-default	active	
1004	fdiinet-default	active	
1005	trnet-default	active	

```

SW1#

```

91.

```

SW1>ena
SW1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#interface g0/1
SW1(config-if)#switchport mode trunk

SW1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

SW1(config-if)#switchport trunk allowed vlan 10,20
SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

```

```

SW3>ena
SW3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW3(config)#interface g0/2
SW3(config-if)#switchport mode trunk

SW3(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

SW3(config-if)#switchport trunk allowed vlan 10,20
SW3(config-if)#end
SW3#
%SYS-5-CONFIG_I: Configured from console by console

```

92.

```
SW3#
SW3#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW3#
```

93.

```
SW2>ena
SW2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW2(config)#interface f0/1
SW2(config-if)#switchport mode access
SW2(config-if)#switchport access vlan 10
SW2(config-if)#exir
^
% Invalid input detected at '^' marker.

SW2(config-if)#exit
SW2(config)#interface f0/2
SW2(config-if)#switchport mode access
SW2(config-if)#switchport access vlan 20
^
% Invalid input detected at '^' marker.

SW2(config-if)#switchport access vlan 20
SW2(config-if)#end
```

R1

```
Router>
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface f0/0
Router(config-if)#ip address 10.10.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface f0/1
Router(config-if)#ip address 10.10.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	10.10.10.1	YES	manual	up	up
FastEthernet0/1	10.10.20.1	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

```
Router#
```

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.20.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/6/16 ms
```

94.

Trunk port

```
SW2(config-if)#switchport mode trunk
SW2(config-if)#switchport trunk allowed vlan 10,20
SW2(config-if)#end
SW2#
```

```
Router(config-subif)#encapsulation dot1
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 10.10.10.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#exit
Router(config)#interface f0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.20, changed state to down
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.20, changed state to up
```

```
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 10.10.20.1 255.255.255.0
% 10.10.20.0 overlaps with FastEthernet0/1
Router(config-subif)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

```
SW2>
SW2>show interface brief
      ^
% Invalid input detected at '^' marker.

SW2>show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	unset	up	up
FastEthernet0/2	unassigned	YES	unset	up	up
FastEthernet0/3	unassigned	YES	unset	down	down
FastEthernet0/4	unassigned	YES	unset	down	down
FastEthernet0/5	unassigned	YES	unset	down	down
FastEthernet0/6	unassigned	YES	unset	down	down
FastEthernet0/7	unassigned	YES	unset	down	down
FastEthernet0/8	unassigned	YES	unset	down	down
FastEthernet0/9	unassigned	YES	unset	down	down
FastEthernet0/10	unassigned	YES	unset	down	down
FastEthernet0/11	unassigned	YES	unset	down	down
FastEthernet0/12	unassigned	YES	unset	down	down
FastEthernet0/13	unassigned	YES	unset	down	down
FastEthernet0/14	unassigned	YES	unset	down	down
FastEthernet0/15	unassigned	YES	unset	down	down
FastEthernet0/16	unassigned	YES	unset	down	down
FastEthernet0/17	unassigned	YES	unset	down	down
FastEthernet0/18	unassigned	YES	unset	down	down
FastEthernet0/19	unassigned	YES	unset	down	down
FastEthernet0/20	unassigned	YES	unset	down	down
FastEthernet0/21	unassigned	YES	unset	down	down

96. Option 3, Layer 3 switching, scales best because it provides high-speed hardware-based inter-VLAN routing, supports many VLANs through SVIs, and avoids the physical-interface limitations of separate router interfaces and the potential trunk bottleneck of router-on-a-stick.

Bonus After all Configurations

